

AI and ML Landscape: Linear Regression (BoomBikes)

Assignment

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Analysis of the categorical variables showed that some categories had a clear influence on bike demand. Variables such as season, weather situation, year, and working day created noticeable differences in average demand, indicating that demand is not driven only by numerical factors. For example, pleasant weather and certain seasons were associated with higher rentals, while poor weather reduced demand. Converting these variables into dummy variables allowed the model to capture these patterns effectively.

2. Why is it important to use `drop_first=True` during dummy variable creation?

Using `drop_first=True` removes one category from each categorical variable, making it the reference category. This prevents the dummy variable trap, where one dummy variable can be perfectly predicted from the others, leading to perfect multicollinearity. It also produces more stable and interpretable regression coefficients.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Based on the pair plot, temperature (`temp`) showed the strongest positive relationship with the target variable (bike demand). As temperature increased, the number of shared bike rentals generally increased as well.

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

After building the model, I checked the key assumptions using diagnostic plots and statistical measures. Residual plots were used to verify linearity and constant variance. A Q-Q plot was examined to check whether the residuals followed a normal distribution. Multicollinearity was assessed using the Variance Inflation Factor (VIF), ensuring that independent variables were not highly correlated. These checks helped confirm that the model was reliable and statistically valid.

5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

The final model indicated that temperature (temp), year (yr), and weather situation (weathersit) were among the most influential variables in explaining bike demand. Temperature had a strong positive impact, while year reflected the increase in demand over time. Weather conditions significantly affected people's willingness to rent bikes.

6. Explain the Linear Regression algorithm in detail.

Linear Regression is a supervised learning algorithm used to predict a continuous value by establishing a relationship between one dependent variable and one or more independent variables. It fits the best possible line by minimizing the squared difference between actual and predicted values. Each feature receives a coefficient that indicates both the direction and magnitude of its influence. Besides prediction, the algorithm also helps identify which variables contribute most to the outcome. Its effectiveness depends on assumptions such as linearity, independence of errors, normality of residuals, homoscedasticity, and low multicollinearity.

7. Explain Anscombe's quartet in detail.

Anscombe's Quartet consists of four datasets that have nearly identical statistical summaries, including mean, variance, correlation, and regression line, yet look very different when plotted. The lesson is that statistics alone can hide important characteristics such as

outliers or non-linear relationships. It highlights why visualizing data is an essential part of data analysis before building a predictive model.

8. What is Pearson's R?

Pearson's R is a statistical measure that indicates the strength and direction of the linear relationship between two numerical variables. Its value ranges from -1 to +1. A value close to +1 represents a strong positive relationship, a value close to -1 indicates a strong negative relationship, and a value near 0 suggests little or no linear relationship. It is commonly used during exploratory analysis to identify variables that are strongly associated with the target variable.

9. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Scaling transforms numerical features to a comparable range so that differences in units or magnitude do not dominate the analysis. It improves numerical stability and makes model interpretation easier. Normalization rescales values to a fixed range, usually 0 to 1, whereas standardization transforms values to have a mean of 0 and a standard deviation of 1. Standardization is generally preferred for regression because it is less sensitive to outliers.

10. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

An infinite VIF occurs when an independent variable can be perfectly predicted using one or more other independent variables. This situation is called perfect multicollinearity. Since the model cannot separate the contribution of those variables, the VIF becomes infinite. A common cause is creating dummy variables without dropping one reference category.

11. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in Linear Regression.

A Q-Q (Quantile-Quantile) plot compares the distribution of model residuals with a theoretical normal distribution. If the points closely follow a straight line, the residuals are approximately normal, indicating that the normality assumption is satisfied. Large deviations from the line suggest skewness, outliers, or other violations. This makes the Q-Q plot an important diagnostic tool for validating a Linear Regression model.